

Submit the following five exercises. You are expected to provide full solutions in a linear and coherent fashion. You are encouraged to work in groups, but your work must be your own and must be self-contained.

Exercise 1.

Let $f(x)$ be a function with $\int_{-1}^1 f(x)dx = -2$ and $\int_1^4 f(x)dx = -3$.

1. Sketch the graph of a function $f(x)$ satisfying the two equalities above.
2. Compute $\int_{-1}^4 5f(x)dx$.
3. Compute $\int_2^2 f(x)dx$, and justify your answer using your sketch above.

Exercise 2.

1. Compute $\int \frac{\cos(x)}{\sin(x)}dx$.
2. Compute $\int e^x \sin(2 + e^x)dx$.

Exercise 3.

Compute $\int (x^2 - 2)\sqrt{x+1}dx$.

Exercise 4.

Compute $\int_{-\pi}^{\pi} x \cos(x^2)dx$.

Exercise 5.

Compute $\int \frac{\sin(x)(\cos(x)+1)}{\cos(x)}dx$.